

Project Overview and Feature-Based Classification Pipeline

The project follows a multi-phase approach, starting from dataset preparation and deep feature extraction, and extending to classical machine learning classifiers and explainability techniques.

In the first phase, we focused on preparing a robust and reproducible image classification pipeline. The PlantVillage dataset, consisting of multiple plant species and disease classes, was organized into training, validation, and test splits. Standard preprocessing and data augmentation techniques were applied to improve generalization and reduce overfitting.

To leverage the representational power of deep convolutional neural networks while avoiding the computational cost of full end-to-end training, we adopted a transfer learning approach. A VGG16 model pretrained on ImageNet was used as a fixed feature extractor. Instead of using the final classification layer, deep features were extracted from the FC6 fully connected layer, which captures high-level semantic information about leaf texture, color, and disease patterns.

These FC6 feature vectors were then used as input to classical machine learning classifiers: Support Vector Machine (SVM) with an RBF kernel and K-Nearest Neighbors (KNN). Experimental results showed that the SVM classifier significantly outperformed KNN, achieving a test accuracy of approximately 96.9% (table 1), which is consistent with results reported in the reference literature (table 2). The slight difference in accuracy can be attributed to variations in data splitting, augmentation strategies, and implementation details. This confirms that combining deep features with non-linear classifiers is an effective strategy for plant disease classification.

Table 1: Comparing Classifiers

Model	Classifier	Test Accuracy
VGG16 + FC6	SVM (RBF)	96.93%
VGG16 + FC6	KNN	89.58%

Table 2: Result of the paper vs our implemetation

Method	Dataset	Accuracy
Paper (VGG16 + FC6 + SVM)	PlantVillage	~97–98%
Our Implementation	PlantVillage	96.93%

While the feature-based classification pipeline achieved strong quantitative performance, it does not provide insight into why the model makes certain predictions. In high-stakes and visually subtle tasks such as plant disease diagnosis, understanding model behavior and confidence is crucial. Therefore, in the next phase of the project, we focused on explainability techniques to analyze the decision-making process of the trained model.

The following sections present the explainability analysis conducted using gradient-based visualization methods and uncertainty estimation techniques, including SmoothGrad, Monte Carlo Dropout, Grad-CAM, Guided Backpropagation, and Guided Grad-CAM.

SmoothGrad and Uncertainty-Based Explainability

We used gradient-based and uncertainty-based explainability methods to understand how stable and reliable the model's predictions are. The chosen model was VGG16 trained on the PlantVillage dataset (with 78% validation accuracy), as the explainability techniques used are architecture-agnostic and can be applied to other CNN models in the same way.

We first prepared the trained VGG16 model for explainability by loading the saved model and ensuring access to its output probabilities. Since the model already contained a dropout layer in the classifier head, it was suitable for both gradient-based saliency analysis and Monte Carlo Dropout.

As a baseline explainability method, we implemented **Vanilla Saliency Maps**, which compute the gradient of the predicted class score with respect to the input image pixels. This method highlights which pixels most strongly influence the model's decision. The resulting saliency maps were a little noisy, but they served as an important reference to be compared with more advanced techniques.

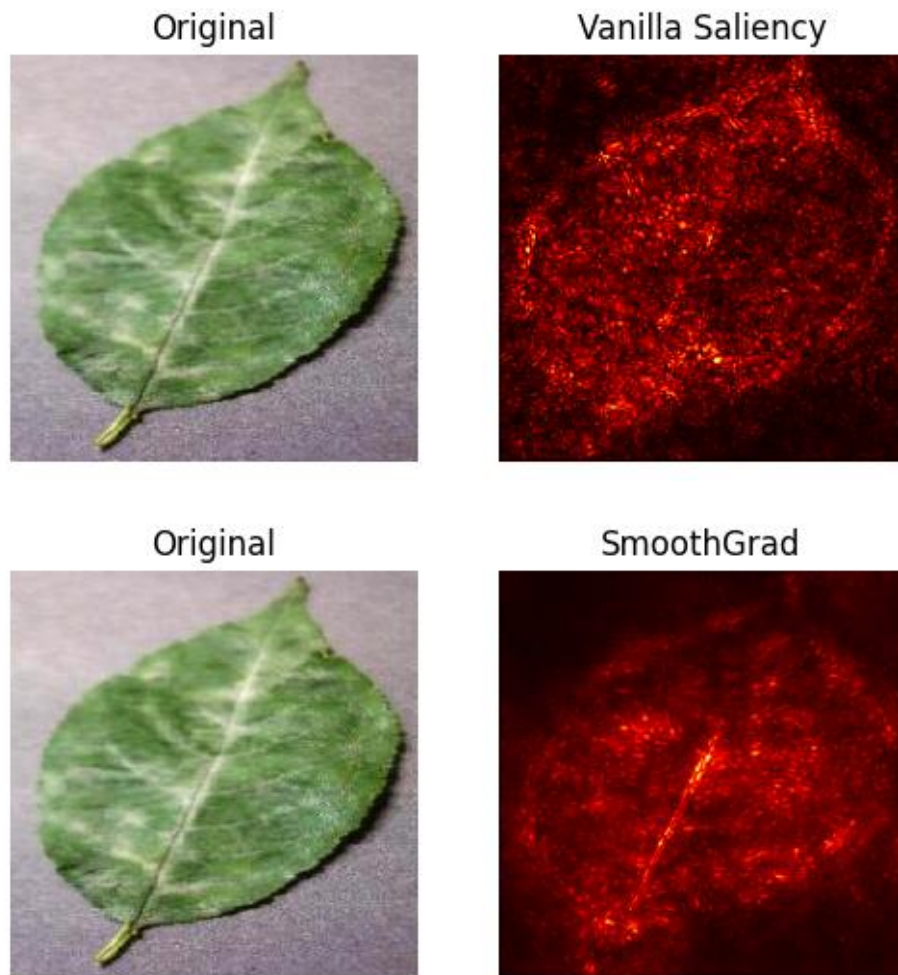
To improve the stability of these explanations, we implemented **SmoothGrad**. In this method, random noise is added to the input image multiple times, gradients are computed for each noisy input, and the results are averaged. This process reduces high-frequency noise and produces smoother saliency maps. Compared to Vanilla Saliency, SmoothGrad highlighted clearer regions on the leaf surface, especially around disease patterns such as spots and texture changes. This confirmed that averaging gradients improves the interpretability of pixel-level explanations.

In addition to saliency-based explanations, we implemented **Monte Carlo Dropout** to estimate predictive uncertainty. By performing multiple forward passes with dropout enabled during inference, we obtained a distribution of predictions for each image. From these predictions, we

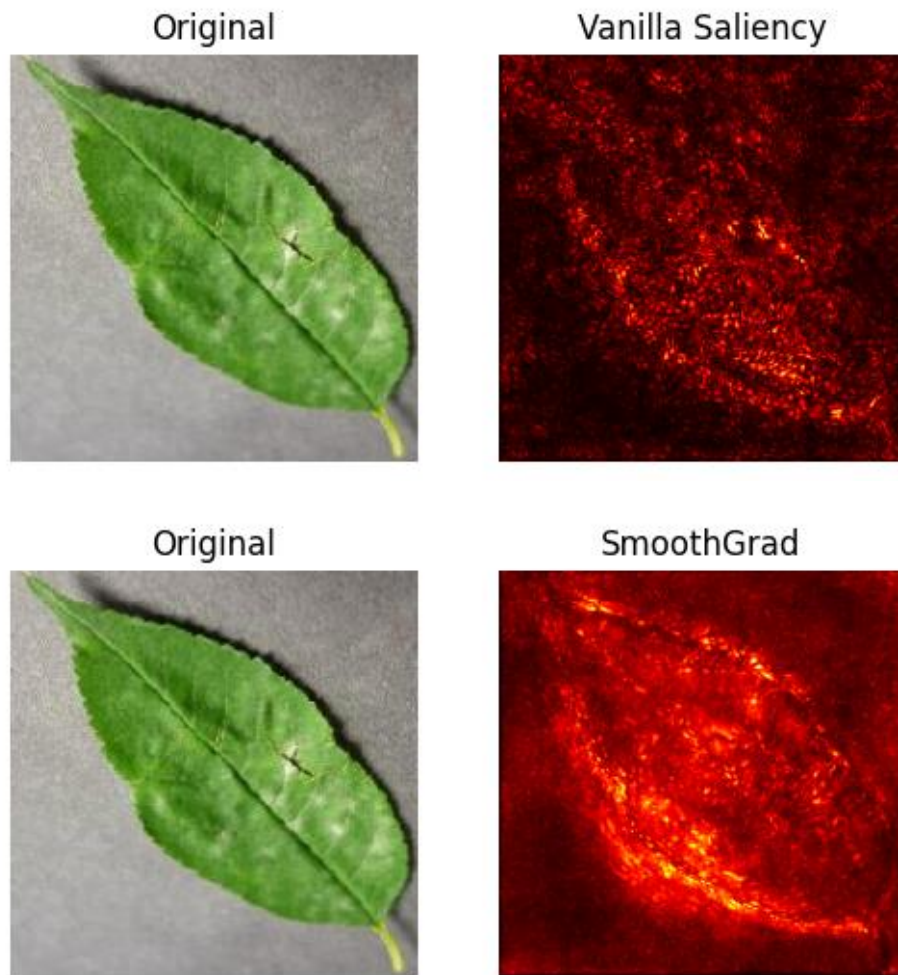
computed the mean class probabilities, per-class variance, and predictive entropy. High variance and entropy values indicated cases where the model was uncertain about its prediction.

We applied these methods to a small subset of test images, including both correctly classified and misclassified examples. Due to computational, we focused on a limited number of images rather than the full dataset. For correctly classified images, saliency maps and SmoothGrad generally focused on relevant leaf regions, while uncertainty values were lower. In contrast, misclassified images often showed higher uncertainty and more diffuse or less meaningful saliency patterns.

An example of a correctly classified image with an uncertainty value of 0.8:



An example of a misclassified image with an uncertainty value of 2.7:



Overall, these experiments show how SmoothGrad improves the robustness of gradient-based explanations and how Monte Carlo Dropout provides valuable insight into model uncertainty. Together, these methods offer complementary perspectives on model behavior: SmoothGrad explains where the model looks, while Monte Carlo Dropout indicates how confident the model is in its decision.

Explainability Analysis Using Grad-CAM, Guided Backpropagation, and Guided Grad-CAM

Then we used Grad-Cam, Guided Backpropagation and Guided Grad-Cam methods, to help visualize which region of the input images contributes most to the model's prediction.

Grad-CAM is first applied to produce heatmaps based on the gradients of the predicted class with respect to the feature maps of the last convolutional layer. These heatmaps provide a coarse spatial representation of the regions that contribute most to the classification. In the visual results, high-importance areas are shown using warmer colors, while regions with little influence appear in cooler tones.

In most examples, the Grad-CAM maps indicate that the model mainly focuses on the leaf area rather than the background. This behavior suggests that the network has learned to identify the relevant object in the image. However, the highlighted regions are often spread over large parts of the leaf and do not always correspond clearly to visible disease symptoms. This may indicate that the model relies more on general leaf characteristics than on specific pathological patterns.

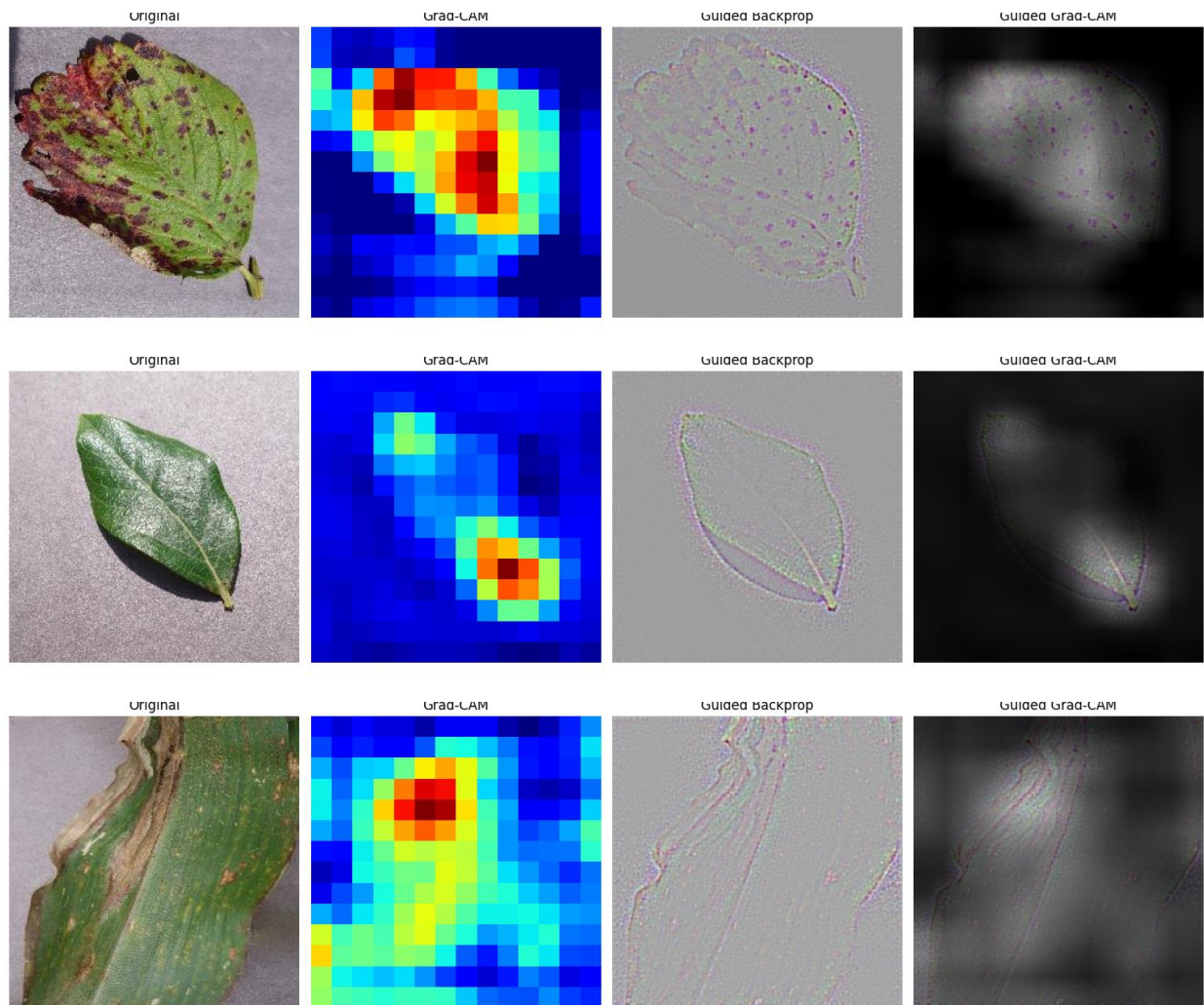
Guided Backpropagation is then used to obtain pixel-level gradient visualizations by modifying the ReLU activations to restrict gradient flow. The resulting images emphasize fine details such as edges, veins, and texture variations. Although these visualizations reveal detailed structural information, they do not directly show which regions are most important for the classification decision.

Guided Grad-CAM combines the localization ability of Grad-CAM with the high-resolution details from Guided Backpropagation. While this method produces sharper and visually more focused maps, the outputs appear very similar across different test images. This similarity suggests that the model responds to comparable visual patterns regardless of the specific input, which may reflect limited class-specific feature discrimination.

The prediction confidence values further support these observations. Images with higher confidence scores tend to exhibit more concentrated activation regions, whereas lower-confidence predictions are associated with more diffuse and less structured attention maps. This indicates a connection between the model's certainty and the clarity of the highlighted regions.

Overall, the explainability results are consistent across images and confirm that the visualization methods are implemented correctly. The main limitations observed in the outputs are therefore more likely related to the trained model itself rather than to the Grad-CAM or Guided Backpropagation implementations. Factors such as dataset composition, training quality, or over-reliance on global features may contribute to these behaviors.

Some samples are shown below:



In summary, the applied explainability techniques provide useful insight into the internal behavior of the model. While the network generally attends to the correct object, it does not always focus on disease-specific areas of the leaf. Improving the training process, increasing data diversity, or refining the model architecture could lead to more meaningful and interpretable visual explanations in future work.

The Grad-CAM results show that the model mainly attends to the leaf regions rather than the background, indicating correct object localization. However, the highlighted areas are often widespread and not clearly focused on disease-specific symptoms, suggesting reliance on global leaf features such as texture and color. The similarity across different Grad-CAM visualizations further indicates limited class-specific feature discrimination, which is likely related to the trained model rather than the explainability method itself.